CONDITIONS ON A POINT IN A FIBRE OF A LOCALLY QUASI-FINITE MORPHISM

THE STACKS PROJECT AUTHORS

Editorial note. No omission markers in-span.

Editorial note. Steps the author leaves to the reader are flagged here for the reviewer: source line(s) 4626 carry an "omitted"/"follows easily"/"immediate from the definitions" phrase. Those sentences are retained verbatim; the displayed argument around them stays inside the counted span.

Allowed standard prerequisites (editorial).

- Fibres of locally quasi-finite morphisms. Morphisms, lemma locally-quasi-finite-fibres (cross-chapter, named). The fibre is a disjoint union of spectra of Artinian local rings.
- Points of the presentation. Properties of Spaces, lemma points-cartesian (cross-chapter, named). The fibre identification.

Source reference. Original lemma 0AC9: sources/stacks-decent-spaces/pinned-original-source.tex, source lines 4442–4442. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 0AC8: sources/stacks-decent-spaces/pinned-original-source.tex, source lines 4392–4392. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 0ABM: focused-seeds/geometry/S9-decent-conditions-on-point-in-fibre-and-qf/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let S be a scheme. Let $f : X \to Y$ be a morphism of algebraic spaces over S which is locally of finite type. Let $x \in |X|$ with image $y \in |Y|$. Let $F = f^{-1}(\{y\})$ with induced topology from $|X|$. Let k be a field and let $\mathrm{Spec}(k) \to Y$ be in the equivalence class defining y. Set $X_k = \mathrm{Spec}(k) \times_Y X$. Let $\tilde{x} \in |X_k|$ map to $x \in |X|$. Consider the following conditions*

(1) $\dim_x(F) = 0$,
(2) *x is isolated in F,*
(3) *x is closed in F and if $x' \rightsquigarrow x$ in F, then $x = x'$,*

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(4) $\dim_{\tilde{x}}(|X_k|) = 0$,
(5) *$\tilde{x}$ is isolated in $|X_k|$*,
(6) *$\tilde{x}$ is closed in $|X_k|$ and if $\tilde{x}' \rightsquigarrow \tilde{x}$ in $|X_k|$, then $\tilde{x} = \tilde{x}'$*,
(7) $\dim_{\tilde{x}}(X_k) = 0$,
(8) *f is quasi-finite at x.*

Then we have

$$(4) \underset{f \text{ decent}}{\Longrightarrow} (5) \Longleftrightarrow (6) \Longleftrightarrow (7) \Longleftrightarrow (8)$$

If Y is decent, then conditions (2) and (3) are equivalent to each other and to conditions (5), (6), (7), and (8). If Y and X are decent, then all conditions are equivalent.

Proof. By Lemma 0AC9 conditions (5), (6), and (7) are equivalent to each other and to the condition that $X_k \to \mathrm{Spec}(k)$ is quasi-finite at $\tilde{x}$. Thus by Morphisms of Spaces, Lemma 0ABM they are also equivalent to (8). If f is decent, then X_k is a decent algebraic space and Lemma 0AC9 shows that (4) implies (5).

If Y is decent, then we can pick a quasi-compact monomorphism $\mathrm{Spec}(k') \to Y$ in the equivalence class of y. In this case Lemma 0AC8 tells us that $|X_{k'}| \to F$ is a homeomorphism. Combined with the arguments given above this implies the remaining statements of the lemma; details omitted. □